

Hypothesis Testing Summary

Objective

To determine whether Technology products generate significantly higher average profits than Furniture products.

Business Hypothesis

Null Hypothesis (H_0)

There is no significant difference in average profit between Technology and Furniture categories.

Alternative Hypothesis (H_1)

Technology products generate significantly higher average profit than Furniture products.

Statistical Test Used

Independent Sample T-Test

Significance Level

$\alpha = 0.05$

Results

P-value: [Insert Your Calculated P-Value]

Decision

If p-value < 0.05, reject the null hypothesis.

Result: [Reject H_0 / Fail to Reject H_0]

Business Conclusion

The statistical analysis indicates that Technology products generate significantly higher average profits than Furniture products.

Recommendation

Increase focus on Technology products through targeted marketing, inventory planning, and promotional campaigns to maximize profitability.